

EE7207 Neural Networks & Deep Learning

Comprehensive CNN: From Foundations to Advanced Tricks

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1. Why CNN? Principles of Efficiency

- **MLP Limitation:** A $200 \times 200 \times 3$ image has $120k$ inputs. One FC layer with 1000 hidden units = 1.2×10^8 parameters. **Result:** High memory usage & Overfitting.
- **CNN Core Pillars:**
 - **Local Connectivity:** Each neuron looks at a small window (*Receptive Field*).
 - **Weight Sharing:** Same filter slides across the image. Feature *A* at $(0, 0)$ is detected the same way at $(10, 10)$.
 - **3D Topology:** Activations are volumes (H, W, C).

Key Term: Translation Equivariance

Weight sharing ensures that if an object moves in the image, its activation map moves accordingly.

2. The Convolutional Layer: Mechanics

- **Hyperparameters:** Filter size (f), Stride (s), Padding (p).
- **Padding:**
 - *Valid:* No padding, output shrinks.
 - *Same:* $p = (f - 1)/2$, output size = input size (if $s = 1$).
- **Filter Depth:** A filter must always have the **same depth** as the input volume (e.g., for RGB, filter is $f \times f \times 3$).

Parameter Calculation Formula

Number of params per layer =
 $[(f \times f \times \text{Input_Depth}) + 1] \times \text{Number_of_Filters}.$

3. Pooling: Reducing Complexity

- **Max Pooling:** Selects the maximum value.
 - *Pro:* Provides **Invariance** to small translations and reduces noise.
- **Average Pooling:** Computes the mean. Usually used at the **Global** level before the final Softmax.
- **Characteristics:**
 - No learnable parameters.
 - Operates on each channel **independently**.

4. Batch Normalization: The Training Accelerator

Problem: *Internal Covariate Shift* – distributions of layer inputs change during training, slowing down convergence.

- **Operation:** Normalizes activations to mean=0, var=1, then scales and shifts.

$$\hat{x} = \frac{x - E[x]}{\sqrt{\text{Var}[x] + \epsilon}}, \quad Y = \gamma \hat{x} + \beta$$

- **Why it works:** Allows higher learning rates and reduces sensitivity to **Weight Initialization**.
- **The (C,) Shape Logic:** Because of weight sharing, we want the same normalization for all spatial locations in a channel to maintain feature consistency.

Question

Output dim = (32, 256, 28, 28). What is the shape of **running mean**?

Correct Choice: (i) (256,)

- **Justification:** BN in CNN is **Channel-wise**. It averages over N, H, W .
- For each of the 256 channels, we store exactly **one** mean value.

5. Classic Architectures: ResNet & MobileNet

- **ResNet (Lecture 6): Identity Shortcut.**
 - Mitigates **Vanishing Gradient**.
 - Gradient can "skip" layers during backprop via $H(x) = F(x) + x$.
- **MobileNet (Lecture 12): Depthwise Separable Conv.**
 - Splits standard conv into:
 - 1 **Depthwise:** $f \times f \times 1$ per channel.
 - 2 **Pointwise:** $1 \times 1 \times C$ to combine channels.
 - **Efficiency:** Reduces computation by $\sim 8\times$ to $9\times$.

6. Training Tricks (Lecture 12)

- **Weight Initialization:**

- **He (MSRA) Initialization:** Best for **ReLU** ($Var = 2/n$).
- **Xavier:** Best for **Tanh/Sigmoid** ($Var = 1/n$).

- **Regularization:**

- **Dropout:** Randomly zero out neurons (train only).
- **Weight Decay ($L2$):** Keeps weights small to prevent overfitting.

7. Transfer Learning Strategy

Scenario	Data & Similarity	Strategy
Case I	Small + Similar	Linear Classifier (on top)
Case II	Large + Similar	Fine-tune full model
Case III	Small + Different	Classifier on earlier layers
Case IV	Large + Different	Fine-tune or Train from scratch

Heuristic: Use earlier layers for different tasks because they detect "low-level" features (edges, blobs) common to all images.

Scenario: Rare cancer (1,000 images), 10:1 imbalance.

- **Mixup:** Mix two images and their labels linearly.
 - *Logic:* Smoothens decision boundaries.
- **CutMix:** Cut a patch from image B and paste onto A.
 - *Logic:* Prevents model from relying on a single visual cue.
- **GANs:** Generate minority class samples.
 - *Logic:* Balances the dataset to 1:1.

Design Choice

For small medical data, choose **CNN** over ViT due to **Inductive Bias** (Locality/Stationarity).